GRAINGUARD · STEM CHECK

GrainGuard STEM check

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GGD-STM-001	Version	0.1
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/grainguard		
Source	docs/01b-stem-check.md @ 5c553ac		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.8 upgrade of a TRL 3 design): 33 claims for option A, 7 for options B and C; seven details fixed under the standing instruction of 2026-10-23 (GGD-DEC-001 D10, D12 to D17); two fixes proposed to Amish (open items 11 and 12); independent review recorded

GrainGuard STEM check

Verdict: GrainGuard's working principle (reading the grain's temperature and equilibrium humidity on a cable, and running the fan only when the plenum air's equilibrium moisture content will cool or dry the grain) sits within known physics, chemistry and biology. Of the 33 claims for option A, the design of record, **19 are within known science, 12 are plausible and wait for TRL 4 data or a purchase check, and 2 fail with a fix proposed to Amish**. Seven details failed and were fixed under Amish's standing instruction of 2026-10-23 (GGD-DEC-001 D10, D12 to D17): the battery's hydrogen had no way out of the sealed box (S18), the mast tube was checked at 235 MPa although its grade is S195T and the mast alone could not carry the 20 W panel with the stay off (S23, S24), nothing said which side of the bin the mast stands on (S15), a fixed 125 kHz LoRa channel is not a legal mode under 47 CFR 15.247 (S20), the plenum probe is unrated below -10 °C (S31, now zeroed against the ambient sensor) and a short in the thin bus cable was protected only by the 5 A battery fuse (S32). Two failures go to Amish because their fixes change settings he decided on 2026-10-02: the battery's electrolyte freezes near -20 °C at the 50 % disconnect, above the -30 °C the design must survive (S17, open item 11), and with a lead-acid charge efficiency added, the 20 W panel refills the battery in 3.5 winter days, not 3 (S14, open item 12). `stem_check` is **open**. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py` , which imports the calculation note GGD-CAL-001 (`docs/04-calcs/sizing.py`).

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` ; S rows are option A, B and C rows are options B and C).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	The reference bin, filled level to the eave, holds about 88.8 t (3,493 bu) of shelled corn	Mathematics	Units and dimensional analysis	Cylinder volume over the 5.2 m from the aeration floor to the eave at 721 kg/m3 bulk density (25.4 kg per 0.0352 m3 bushel)	claimed 88.8 t, estimate 88.8 t (ratio 1.00)	Within known science	GGD-CAL-001 A2
S2	The fan's own power warms the air by up to about 1.0 °C at 0.40 kW input (an upper bound: a motor outside the air stream adds less)	Physics	Thermodynamics	Steady-flow energy balance, dT = P / (rho Q cp), all motor and fan losses into the air stream	claimed 1 K, estimate 1 K (ratio 1.00)	Within known science	GGD-CAL-001 A7; measured by the plenum probe (B10)
S3	Static pressure is about 139 Pa (0.56 in wc) at 0.2 cfm/bu through 5.2 m of corn	Physics	Fluid mechanics	Shedd's equation for shelled corn (ASABE D272), packing factor 1.5 and 50 Pa of duct and floor loss; estimate without the packing factor	claimed 139 Pa, estimate 109 Pa (ratio 1.27)	Within known science	GGD-CAL-001 A4
S4	A cooling front passes through the bin in 75 to 119 h of fan time	Physics	Heat transfer	Sensible heat balance (grain heat capacity over the air's heat capacity rate) against the extension rule of 15 h divided by cfm/bu (University of Minnesota)	claimed 119 h, estimate 75 h (ratio 1.59)	Within known science	GGD-CAL-001 A8
S5	Air at 10 °C and 70 % RH holds shelled corn at about 15.7 % wet basis (its equilibrium	Chemistry	Physical chemistry (equilibria)	ASABE D245.6 modified Henderson equation with the shelled-corn constants; the basis of	15.70 % wet basis from the equation	Within known science	GGD-CAL-001 B1

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	moisture content)			extension aeration charts			
S6	Each 1 °C of fan heat lowers the plenum air's EMC by about one point at 10 °C and 70 % RH (about 1.5 points at 0 °C and 80 %, over 2 points in cold air near 90 %)	Chemistry	Physical chemistry (equilibria)	Sensible heating at constant humidity ratio (Tetens saturation pressure), then the EMC equation	claimed 0.97 points, estimate 0.972 points (ratio 1.00)	Within known science	GGD-CAL-001 B1
S7	With typical sensor tolerance the moisture estimate moves by up to 0.23 points (SHT45) and 0.42 points (SHT40) over the specified 20 % to 75 % RH (0.81 at the SHT40 maximum tolerance; 0.83 for the SHT40 at 5 °C and 92 %, outside the range)	Mathematics	Statistics and uncertainty	Propagation of the Sensirion typical tolerances (±1.0 and ±1.8 % RH, ±0.1 and ±0.2 °C) through the EMC equation	0.42 points <= 0.8 points	Within known science	GGD-CAL-001 B5
S8	The moisture estimate meets ±0.8 points once the equation's own fit error, sorption hysteresis, sensor drift and membrane lag are added	Chemistry	Analytical chemistry	USDA ARS work on RH-based grain moisture prediction reports errors of the order of one point; hysteresis between wetting and drying is about the same size	Not closable by calculation; comparison with oven or meter samples at TRL 4 (PR-002 at risk)	Plausible, to confirm at TRL 3	Open
S9	Interstitial air comes to	Physics	Thermodynamics	Moisture exchange	Time to equilibrium in	Plausible, to	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	equilibrium with the kernels within about 6 h after the fan stops, so a pod reading is a grain reading			between kernels and still interstitial air is fast at the kernel surface; the PTFE membrane adds a lag	the pod to measure at TRL 4 (GGD-PRC-001 open question)	confirm at TRL 3	
S10	A temperature sensor within about 0.5 m of a developing hot spot detects it; one center cable watches 3.3 % of the cross-section	Biology	Microbiology	Ileleji et al., Applied Engineering in Agriculture 22(2), Purdue (hot spot detection radius)	3.3 % of 23.7 m2 of cross-section within 0.5 m of the cable (about 2.5 % of the grain volume if each pod watches a 0.5 m sphere); the rest is unwatched (PR-006 at risk, a coverage limit, not a science failure); the 0.5 m radius rests on one study	Within known science	GGD-CAL-001 C2
S11	Running fans when outside air is cooler than the grain holds it cool and uniform, which slows insects and mould	Biology	Ecology and agronomy	Oklahoma State University BAE-1116 and University of Minnesota Extension aeration guidance	Practice the control rule follows (GGD-DDR-001 D3)	Within known science	GGD-PRC-001, control rule
S12	A heating alarm reaches the farmhouse within 15 min	Engineering and technology	Control	10 min sampling, one LoRa packet under 0.4 s and three retries 60 s apart	13 min <= 15 min	Within known science	GGD-CAL-001 C4
S13	The battery lasts 7.7 days with no sun at -20 °C with the fan relay held on	Engineering and technology	Power and link budgets	Energy balance: 7 Ah x 12 V, 50 % usable to the disconnect, 60 % of that at -20 °C, against 3.28 Wh per day, of which 0.86 Wh is	7.7 days against 5; rests on a relay input of 3 mA or less (a coil relay held on draws 30 to 70 mA and would give 1 to 2 days):	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				the relay input at 3 mA	to confirm from the input module's datasheet at purchase (GGD-DEC-001 confirm list)		
S14	The 20 W panel refills the battery from half charge in 3 winter days at 1.5 peak sun hours	Physics	Optics and radiation	Solar yield = rated power x peak sun hours x 0.6 derating x 0.85 lead-acid charge efficiency, less the daily load (the charge efficiency was missing until the independent review)	3.5 days against 3 (2.9 days without the charge efficiency); a panel of about 23 W is needed; fix proposed to Amish because the 20 W panel was his decision (GGD-DEC-001 open item 12); 1.5 peak sun hours to confirm from NREL data for the host site	Fails, fix proposed	Open
S15	The panel, 0.7 m from a 5.6 m wall and facing away from it, is in sun all winter	Physics	Optics and radiation	Solar geometry: noon elevation 90 - latitude - 23.44 degrees at the winter solstice	Noon sun 24.6 degrees at 42 °N, wall shadow 12.3 m: a mast on the north side is shaded all day, so the design needed a siting rule; with the mast within 30 degrees of due south of the bin axis the bin, seen from the mast, fills azimuths 277 to 23 degrees while the winter sun stays between about 126 and 234 degrees, so the bin never shades the panel at any latitude in the Corn Belt;	Within known science	GGD-DEC-001 D13 (fix: siting rule)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					other obstructions are a site check		
S16	With temperature compensation clamped at 15.0 V the battery bus never exceeds 15 V (the panel circuit, up to about 25 V open circuit in the cold, is covered separately in PR-012)	Chemistry	Electrochemistry and batteries	AGM absorption 14.4 V at 25 °C, -30 mV/K; 15.75 V unclamped at -20 °C; the clamp's own tolerance to confirm at purchase	15 V <= 15 V	Within known science	GGD-CAL-001 D8; GGD-DEC-001 D14 (PR-012 wording)
S17	The AGM battery survives -30 °C with the low-voltage disconnect at the maker's 50 % figure	Chemistry	Electrochemistry and batteries	Electrolyte freezing point against state of charge, BCI Battery Service Manual table (Trojan): 62 % freezes at -27 °C, 40 % at -15 °C	50 % freezes near -20 °C, above the -30 °C of PR-009; the 70 % figure freezes near -36 °C and leaves 4.6 to 7.7 days of autonomy, depending on whether the cold derating applies at this low draw; fix proposed to Amish (GGD-DEC-001 open item 11)	Fails, fix proposed	Open
S18	Battery hydrogen leaves the sealed enclosure by natural draught	Chemistry	Electrochemistry and batteries	IEC 62485-2 natural ventilation: Q = 0.05 n Igas C 1e-3 m3/h (valve-regulated cells, boost 8 mA/Ah), opening A = 28 Q cm2 in and out; a membrane breather has no open area	0.79 cm2 >= 0.47 cm2	Within known science	GGD-DEC-001 D10 (fix: two M16 labyrinth vents, in the bottom and at the top of the side wall; the effective area of the vent maker's

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
							labyrinth to confirm)
S19	The LoRa link has 10 dB or more of margin at 1 km past one farm building	Physics	Electromagnetism and circuits	Plane-earth path loss with the modelled antenna heights, 20 dB for one building (an assumption), sensitivity -126 dBm at SF10 and 500 kHz	10.9 dB on paper; bare-ground propagation ignores a summer corn crop round antennas 2 m up and the bin's own shadow, so a site survey decides (TRL 4)	Plausible, to confirm at TRL 3	Open
S20	The radio operates legally in the US 902 to 928 MHz band	Engineering and technology	Standards practice	47 CFR 15.247: a fixed 125 kHz channel is neither a hopping system (50 channels, 0.4 s dwell) nor a digital modulation system (6 dB bandwidth 500 kHz or more), so the radio runs at SF10 with a 500 kHz bandwidth under 15.247(a)(2); check of its power spectral density against the 8 dBm in 3 kHz limit	-8.2 dBm/3 kHz <= 8 dBm/3 kHz	Within known science	GGD-DEC-001 D15 (fix: 500 kHz bandwidth); GGD-CAL-001 E5, E6
S21	The 6 mm rope at its clipped eye holds 4 times the 2.5 kN design pull-down	Physics	Strength of materials	20 kN minimum breaking load x 0.8 termination efficiency of wire rope clips against R10's factor of 4; 3.4 times the 4.7 kN measured on the largest commercial cable (Illinois, ASABE)	16 kN >= 10 kN	Within known science	GGD-CAL-001 F3

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S22	Grain pulls the thin cable down with about 2.1 kN while the bin is unloaded	Physics	Mechanics and statics	Janssen pressure with friction on the rope and the pods' projected area, x 2 for unloading; forces up to 4.7 kN were measured on the largest commercial cable (Illinois, ASABE)	Estimate 2.08 kN; no published measurement for a 6 mm cable with pods, so unverified; the rope holds even the 4.7 kN measured (S21)	Plausible, to confirm at TRL 3	Open
S23	With the stay fitted, the DN25 mast carries the 20 W panel and the shield in a 45 m/s gust	Physics	Strength of materials	Cantilever above the stay, drag coefficient 1.2 on the panel; S195T tube (EN 10255), yield 195 MPa, factor 1.5	73 MPa <= 130 MPa	Within known science	GGD-CAL-001 F4; GGD-DEC-001 D12 (yield of the tube corrected from 235 MPa)
S24	With the stay off for service, the mast alone carries the panel	Physics	Strength of materials	Cantilever from the base, S195T tube; 257 MPa at 45 m/s (fails), so the stay comes off only in gusts under 30 m/s	114 MPa <= 130 MPa	Within known science	GGD-DEC-001 D12 (fix: 30 m/s service rule, closes open item 9)
S25	The 40 x 40 x 4 stay angle carries the mast's prop force	Physics	Strength of materials	Prop force with the welded base as a fixed end (R = P (1 + 1.5 c / a)); guided cantilever along the wall, S235, factor 1.5; Euler buckling toward the wall	76 MPa <= 157 MPa	Within known science	GGD-CAL-001 F5
S26	Four M10 concrete anchors hold the base plate (1.15 kN on the worst anchor at the 30 m/s service limit,	Engineering and technology	Standards practice	Anchor tension capacity depends on the anchor's assessment, the concrete grade and the edge distance	To confirm against the chosen anchor maker's data and the pad's concrete when the site is known	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	2.59 kN at 45 m/s, wind on the diagonal)						
S27	The relay kit isolates the 12 V controller from the starter's mains by at least 2.5 kV	Physics	Electromagnetism and circuits	Optically isolated relay input modules are specified to this class of isolation by their makers (datasheet item at purchase)	Specification of BOM line 11; a purchase check, not a calculation	Plausible, to confirm at TRL 3	Open
S28	Conformal coating protects the pod boards from phosphine fumigation	Chemistry	Inorganic chemistry	Phosphine corrodes copper and silver (FAO fumigation manual); coatings slow it, but their life in phosphine is not published for this use	A coated test board through one fumigation at TRL 4 (decided 2026-10-02)	Plausible, to confirm at TRL 3	Open
S29	The 12 V, few-milliamp pods and cable cannot ignite grain dust in the headspace, electrically or by static discharge from plastic pods in flowing grain	Chemistry	Physical chemistry (kinetics)	Grain dust is a combustible dust (NFPA 70 Article 502, Class II); intrinsic safety needs an assessed energy limit; plastic parts in flowing grain can charge	To assess with a hazardous-locations reviewer (REG A2); the rope and cable shield are bonded to the bin	Plausible, to confirm at TRL 3	Open
S30	The six-plate radiation shield keeps the ambient sensor within ±0.3 °C of the air	Physics	Heat transfer	Passive multi-plate shields show radiation errors of tenths of a degree to over 1 °C in sun and low wind	Shield error to measure at TRL 4 (PR-003)	Plausible, to confirm at TRL 3	Open
S31	The plenum probe knows the fan heat within about ±0.5 °C below -10 °C, where the DS18B20's	Engineering and technology	Sensing resolution	DS18B20 rated ±0.5 °C from -10 to +85 °C only; most of its error is an offset, which the controller now	Residual error after zeroing to measure at TRL 4	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	rated ±0.5 °C no longer applies			zeroes against the ambient sensor whenever the fan has been off for an hour (GGD-DEC-001 D16)			
S32	A short in the bus cable inside the bin clears before the cable heats	Physics	Electromagnetism and circuits	A 0.5 A fuse on the bus cable's 12 V feed (GGD-DEC-001 D17) against the current a 0.34 mm2 core carries without heating (about 3 A in free air, a common rating to confirm on the cable's datasheet)	0.5 A <= 3 A	Within known science	GGD-DEC-001 D17 (fix: bus fuse; the 5 A battery fuse alone did not protect the 0.34 mm2 cores)
S33	Snow sliding off the roof does not strike the mast's panel and box 0.7 m from the wall	Physics	Mechanics and statics	Snow leaving a 30 degree roof at the eave falls close to the wall; the panel top is at about 2.4 m under a 5.6 m eave	Site-dependent; to assess at the host bin (a snow guard or a larger offset would need Amish's decision)	Plausible, to confirm at TRL 3	Open
B1	Option B: during a fan run the air leaving the grain carries the top layer's temperature and equilibrium RH	Physics	Thermodynamics	Psychrometrics of the air leaving the grain; the cooling front reaches the surface last (Oklahoma State University)	Sees the front arrive at the top; no profile below it	Within known science	GGD-OPT-001
B2	Option B: headspace CO2 rises when grain begins to spoil, before a distant cable sees heat	Biology	Microbiology	Purdue field trial at The Andersons, Delphi, Indiana (2005 to 2006): headspace CO2 rose from 500 to 5,000 ppm as a tank spoiled	Within known science; thresholds per crop to set with the extension specialist	Within known science	GGD-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
B3	Option B: headspace readings stand in for grain moisture and temperature	Chemistry	Analytical chemistry	The headspace mixes air from the whole surface and the roof; it gives no level-by-level profile	PR-001 not met; moisture only as a bin-top average	Plausible, to confirm at TRL 3	Open
B4	Option B: two 0.3 m wall arms on the stiffener seam bolts carry the wall unit in a 45 m/s gust	Physics	Strength of materials	Cantilever of the arms and the seam bolts in shear and tension; the bin maker's rating of the stiffener bolts	Calculation and the bin maker's rating if B is chosen	Plausible, to confirm at TRL 3	Open
C1	Option C: outside-air EMC with the measured fan heat decides when aeration helps	Chemistry	Physical chemistry (equilibria)	As S5 and S6	Within known science	Within known science	GGD-OPT-001
C2	Option C: counting fan hours against a cooling-front model tells when a front has passed	Physics	Heat transfer	Front time is set by airflow, which varies with fill depth and fines; with no grain sensor the model is never checked	Works as a schedule, not as a measurement; no heating alarm (PR-006 not met)	Plausible, to confirm at TRL 3	Open
C3	Option C: the relay kit's 24 V supply powers the controller, so no panel or battery is needed	Engineering and technology	Power and link budgets	A 24 V DIN-rail supply of 0.5 A or more against a controller of under 1 W	Within known science; it stops when the starter's supply is off	Within known science	GGD-OPT-001

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. GrainGuard is already at TRL 3, so the plausible rows that cannot be closed by calculation (S8, S9, S13, S19, S22, S26 to S31, S33) are confirmed by measurement at TRL 4 or by a purchase check; they keep `stem_check` open but do not change the TRL (STANDARDS section 20).

Fixes made under the standing instruction

Table 2. Details that failed and the fix made (GGD-DEC-001, decided under Amish's standing instruction of 2026-10-23; Amish can reverse each).

Row	What failed	Fix	Result
S18	The AGM battery sits in a sealed IP66 box with one M12 membrane breather. A valve-regulated battery gives off hydrogen when it is charged hard (15.0 V in the cold), and a membrane breather has no open area, so by the IEC 62485-2 method the box had no natural ventilation	D10: two M16 labyrinth vents with a clear bore of 10 mm, one in the bottom face (the old breather's place) and one at the top of the left side wall, its locknut against the inside of the top so no gas pocket is left; USD 4.00 added to BOM line 4 (an estimate)	0.79 cm² open area each against 0.47 cm² needed (1.7 times, before the labyrinth's own loss, to confirm); the box's ingress rating becomes the vents' (IP55 or better), recorded in PR-009
S23	The mast checks used 235 MPa, but 33.7 x 3.2 mm galvanized tube is EN 10255 medium series in S195T, with a minimum yield of 195 MPa	D12: the calculation uses 195 MPa for the tube	Stay fitted: 73 MPa, 2.7 times yield at a 45 m/s gust (was 3.2)
S24	With the stay off for service, the mast alone sees 257 MPa at a 45 m/s gust with the 20 W panel, over yield (open decision 9 since 2026-10-02)	D12: the stay comes off only in gusts under 30 m/s (65 mph), on a calm day, and is refitted before leaving; the rule is in the build plan. A DN32 mast would not reach the 1.5 times margin either (about 140 MPa against 130 MPa)	114 MPa at 30 m/s, 1.7 times yield; worst anchor 1.15 kN; open decision 9 is closed by this fix
S15	The panel faces away from the bin, 0.7 m from a 5.6 m wall, but nothing fixed where round the bin the mast stands. At the winter solstice the noon sun is 24.6 degrees up at 42 °N and the wall's shadow is 12.3 m long, so a mast on the north side would be in shade all day	D13: the mast stands on the south side of the bin, within 30 degrees of due south of the bin axis (north side in the southern hemisphere)	The bin never shades the panel in winter (the bin fills azimuths 277 to 23 degrees as seen from the mast; the winter sun stays between about 126 and 234 degrees)
S20	The radio was planned for one fixed 125 kHz channel, relying on the 0.4 s dwell limit. Under 47 CFR 15.247 that limit belongs to hopping systems of 50 channels or more; a fixed narrow channel is neither a hopping nor a digital modulation system (found by the independent review)	D15: the radio runs at SF10 with a 500 kHz bandwidth on one channel, the digital modulation route of 15.247(a)(2), at both ends; firmware setting only	Packet 93 ms; spectral density -8.2 dBm in 3 kHz against 8 dBm; the link margin falls from 16.9 dB to 10.9 dB at 1 km (PR-007 still met on paper; S19 plausible until a site survey)
S31	The plenum probe's ±0.5 °C rating stops at -10 °C, but the fan runs colder; each degree of error is about a point of EMC (independent review)	D16: the controller zeroes the probe against the ambient sensor whenever the fan has been off for an hour, removing most of its offset; firmware only	Residual error to measure at TRL 4 (plausible)
S32	The bus cable's 0.34 mm² cores were protected only by the 5 A battery fuse, so a short inside the bin could heat the cable in the grain dust (independent review, missing load case)	D17: a 0.5 A fuse on the bus cable's 12 V feed at the surge strip (BOM line 16, no price change)	Clears a cable short at a sixth of the cores' rating

None of these fixes changes the working principle, the pitch or the scope. The D3 model of kit 1.9 also found five fit details that the drawings had left open (jumper cables crossing at each pod, the battery shelf screw heads under the battery, and fixings for the charge controller, radio board and fuse rail); they are recorded in GGD-DEC-001 D11 as design-for-construction details. D14 corrects the wording of PR-012: the 15 V limit applies to the battery bus, the controller's outputs, the bus cable and the pods; the panel circuit runs up to about 25 V open circuit in the cold, still extra-low voltage.

Fixes proposed to Amish

Two rows are **Fails, fix proposed** and go to Amish rather than being fixed here, because each fix changes a setting he decided on 2026-10-02.

- **S17, battery freezing (GGD-DEC-001 open item 11).** By the BCI freezing table the electrolyte freezes near -20 °C at 50 % charge, near -30 °C at 65 % and near -36 °C at 70 %. The usable energy at those settings gives 7.7, 5.4 and 4.6 days with no sun if the 60 % cold derating of section D applies, or 12.8, 9.0 and 7.7 days if it does not at the station's very low draw (about C/600). The recommendation is the 70 % figure: it leaves a 6 K margin below -30 °C, and its autonomy is met on the low-draw basis and short on the conservative one, which a cold-chamber test at TRL 4 settles. An empty battery stops the fan (the safe state); a frozen one cracks.
- **S14, winter recovery (GGD-DEC-001 open item 12).** With a lead-acid charge efficiency of 85 %, which the 60 % derating did not include, the 20 W panel refills from half charge in 3.5 days at 1.5 peak sun hours; about 23 W is needed for 3 days. The recommendation is a 25 W panel (2.7 days), with the panel bracket and the mast checks rerun for it (the stay-off service limit would fall to about 27 m/s); the alternative is to restate the target as 3.5 days.

Fields not relevant

None. Every field is touched: physics (heat balance, fluid flow through grain, solar geometry, radio propagation, mast and rope strength), chemistry (sorption equilibrium, battery electrochemistry, phosphine and dust), biology (spoilage, hot spots, insects), mathematics (units, uncertainty propagation) and engineering (power and link budgets, control, radio rules, anchors).

Concept options

- **Option A, center sensor cable with EMC fan control:** rows S1 to S33. Within known science; two fixes proposed to Amish (S14, S17).
- **Option B, headspace sensor head with CO_2 watch:** within known science (B1, B2): the air leaving the grain carries the top layer's state during a fan run, and headspace CO_2 rises when grain spoils. It cannot give a level-by-level profile (B3, PR-001 not met), and its wall arms need a calculation and the bin maker's bolt rating if it is chosen (B4). It shares rows S5, S6, S12 to S20, S27, S31 and S32 with option A.
- **Option C, fan-side weather controller:** within known science (C1, C3) as a schedule; with no grain sensor its cooling-front model is never checked and it gives no heating alarm (C2, PR-006 not met). It shares S2, S5, S6, S11, S19, S20, S27 and S31 with option A, and has no battery or panel, so S13, S14, S16, S17 and S18 do not apply to it.

No option has a **Fails** row, so none is dropped.

Independent review

A second agent, a separate headless Claude session given only this STEM check, the calculation note GGD-CAL-001, the product requirements GGD-PRS-001 and `sizing.py` (and none of the author's reasoning), was asked on 2026-10-24 to refute the headline results and every "Within known science" verdict. It recomputed S1 to S7, S13 to S16, S19 to S21, S23 and S24 and reproduced every number to the stated digits; its findings are in the assumptions and the verdicts. What it found and what changed:

Table 3. Independent review findings.

Row	Finding	Change
S1, S4, S5, S12, S22	Upheld	None
S2	Upheld as an upper bound (a motor outside the air stream adds less heat)	Claim worded as an upper bound
S3	Weakened: the reviewer recalled other Shedd constants for corn (unverified) that would double the pressure	Kept: the constants used (a 2.07e4, b 30.4) are those of ASABE D272 for shelled corn as cited in GGD-CAL-001; the static pressure does not enter any requirement
S6	Weakened: about one point per degree holds at 10 °C and 70 % RH; about 1.5 at 0 °C and 80 %, over 2 near 90 %	Claim gives the range; the plenum probe matters most in cold, damp air
S7	Weakened: typical tolerance, and the range stops at 75 % RH; the SHT40 reaches 0.83 points at 5 °C and 92 %	Claim reworded ("up to", typical, inside the specified range) with the out-of-range figure
S8, PR-002	Weakened: the cited USDA ARS error of about one point exceeds 0.8	PR-002 stays at risk; the comparison with oven samples at TRL 4 decides
S10	Weakened: six 0.5 m spheres cover about 2.5 % of the volume, not 3.3 %	Both figures stated; the 0.5 m radius rests on one study
S13	Weakened: rests on a relay input of 3 mA; a coil relay held on would give 1 to 2 days	Now plausible, closed by the input module's datasheet at purchase (confirm list)
S13, S17, G3	Weakened: the 50 % disconnect and the 60 % cold derating are applied together, while freezing follows the chemical state of charge	G3 gives both bases; open item 11 now recommends the 70 % figure (7.7 days on the low-draw basis, 4.6 on the conservative)
S14	Refuted: no charge efficiency; at 85 % the refill takes about 3.4 to 3.5 days	Now **Fails, fix proposed** (open item 12); 1.5 peak sun hours to confirm from NREL data for the site
S15	Upheld at 42 °N; the reviewer asked for a tighter angle and other latitudes	Kept at 30 degrees: the bin's shading sector (277 to 23 degrees from the mast) never meets the winter sun (126 to 234 degrees) at any Corn Belt latitude; other obstructions added to the site checks
S16	Weakened: 15 against 15 has no margin, and the panel circuit exceeds 15 V	Claim limited to the battery bus; PR-012 reworded (D14); clamp tolerance a purchase check
S18	Upheld; the high vent left a gas pocket and the labyrinth's own area is less than its bore	Upper vent moved to the top of the side wall (D10); effective area to confirm
S19	Weakened: the building loss is assumed, and a summer corn crop and the bin itself are not in the model	Now plausible; site survey at TRL 4
S20	Refuted: a fixed 125 kHz channel is not covered by the hopping rule	Fixed by D15 (500 kHz digital modulation); the margin falls to 10.9 dB
S21	Weakened: the clipped eye holds about 80 % of the rope's breaking load	Claim restated on the design force: 16 kN at the eye, 6.4 times 2.5 kN; 3.4 times the 4.7 kN measured on a larger cable
S23, S25		

Row	Finding	Change
	Upheld for the mast; the stay's prop force assumed a pinned base	F5 redone with the welded base as fixed: 397 N, 76 MPa (was 67)
S24	Weakened: a weather rule is administrative	Kept as the fix (D12); the reviewer would rate it plausible; the build plan makes refitting the stay a hold point before leaving
S26	Weakened: wind on the plate's diagonal loads one anchor 1.41 times more	F6 uses the diagonal: 1.15 kN at 30 m/s, 2.59 kN at 45 m/s
S27	Weakened: a purchase-time specification, not science	Now plausible
Plenum probe below -10 °C	Missing claim	S31 added; probe zeroed against the ambient sensor (D16)
Missing load cases	Static discharge in flowing grain; a short in the bus cable; roof snow falling on the mast; wall bracket and stiffener bolt capacity; torsion from the panel's offset and vortex shedding; filling and settling loads on the rope; cold limits of the parts	S29 extended to static; S32 added and fixed (D17); S33 added (plausible, site check); the bracket and stiffener bolts added to the confirm list; the rest recorded in GGD-RSK-001 and the review note for TRL 4

The reviewer's recount, on the version it saw, was about 15 within, 14 plausible and 1 fail. The present count (19 within, 12 plausible, 2 fails with a fix proposed) differs because S20 and S32 were fixed after the review and now close within known science, S14 moved to a proposed fix, and S15, S21 and S24 stay within on calculation with the reviewer's concerns recorded above.

Sources

- ASABE D245.6, *Moisture Relationships of Plant-Based Agricultural Products* (modified Henderson constants for shelled corn), as used in GGD-CAL-001.
- ASABE D272, *Resistance to Airflow of Grains, Seeds, Other Agricultural Products, and Perforated Metal Sheets* (Shedd's equation), as used in GGD-CAL-001.
- Oklahoma State University Extension, BAE-1116, "Aeration Management: Knowing When to Run Aeration Fans," https://extension.okstate.edu/fact-sheets/aeration-management-knowing-when-to-run-aeration-fans.html, accessed 2026-10-24.
- University of Minnesota Extension, "Managing stored grain with aeration," https://extension.umn.edu/corn-harvest/managing-stored-grain-aeration, accessed 2026-10-24.
- K. E. Ileleji et al., hot spot detection in stored grain, *Applied Engineering in Agriculture* 22(2), Purdue University, https://engineering.purdue.edu/ABE/people/Papers/klein.ileleji.1/hotspot, accessed 2026-10-24.
- USDA ARS, accuracy of grain moisture content prediction from air relative humidity, https://www.ars.usda.gov/ARSUserFiles/30200525/362AccuracyGrainMoistureContentPrediction.pdf, accessed 2026-10-24.
- Illinois Experts, "Forces on monitoring cables during grain bin filling and emptying" (ASABE), https://experts.illinois.edu/en/publications/forces-on-monitoring-cables-during-grain-bin-filling-and-emptying/, accessed 2026-10-24.

- Parker Steel, declaration of performance for S195T gas and water tube to EN 10255 (minimum yield 195 MPa), https://parkersteel.co.uk/webshared/media/pdf/dop/PS-DOP028-S195T-GAS-WATER.pdf, accessed 2026-10-24.
- Trojan Battery, *Deep Cycle Battery Storage Guide*, electrolyte freezing point at states of charge from the BCI *Battery Service Manual* (1995), https://ressupply.com/documents/trojan_battery/Deep_Cycle_Battery_Storage_Guide.pdf, accessed 2026-10-24.
- INOTEC, "Venting and aerating batteries" (EN 50272-2 ventilation formula Q = 0.05 n Igas C 10^{-3} m^3/h and opening A = 28 Q), https://www.inotec-licht.de/en/practice/venting-and-aerating-batteries/, accessed 2026-10-24. EN 50272-2 is replaced by IEC 62485-2; the boost value of Igas for valve-regulated cells (8 mA/Ah) is taken from that standard's table and is to be confirmed from its text.
- 47 CFR 15.247, operation within the bands 902 to 928 MHz (0.4 s average occupancy in 20 s for hopping channels under 250 kHz), Legal Information Institute, https://www.law.cornell.edu/cfr/text/47/15.247, accessed 2026-10-24.
- World Grain, "Early spoilage detection" (Purdue field trial of headspace CO_2 monitoring at The Andersons, Delphi, Indiana, 2005 to 2006), https://world-grain.com/articles/10162-early-spoilage-detection, accessed 2026-10-24.
- FAO, *Manual of fumigation for insect control* (phosphine and metals), https://www.fao.org/4/x5042e/x5042e0a.htm, accessed 2026-10-24.
- EC&M, "NEC Article 502 summary" (Class II locations, combustible dust), https://www.ecmweb.com/national-electrical-code/article/21281848/nec-article-502-summary, accessed 2026-10-24.
- Sensirion SHT45 and SHT40 datasheets, https://sensirion.com/products/catalog/SHT45 and https://sensirion.com/products/catalog/SHT40, accessed 2026-10-24.